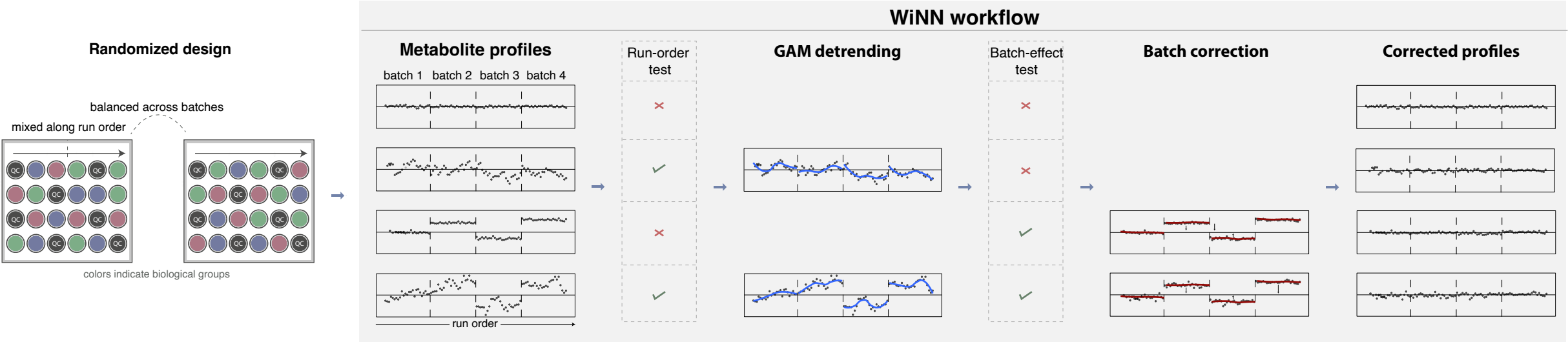
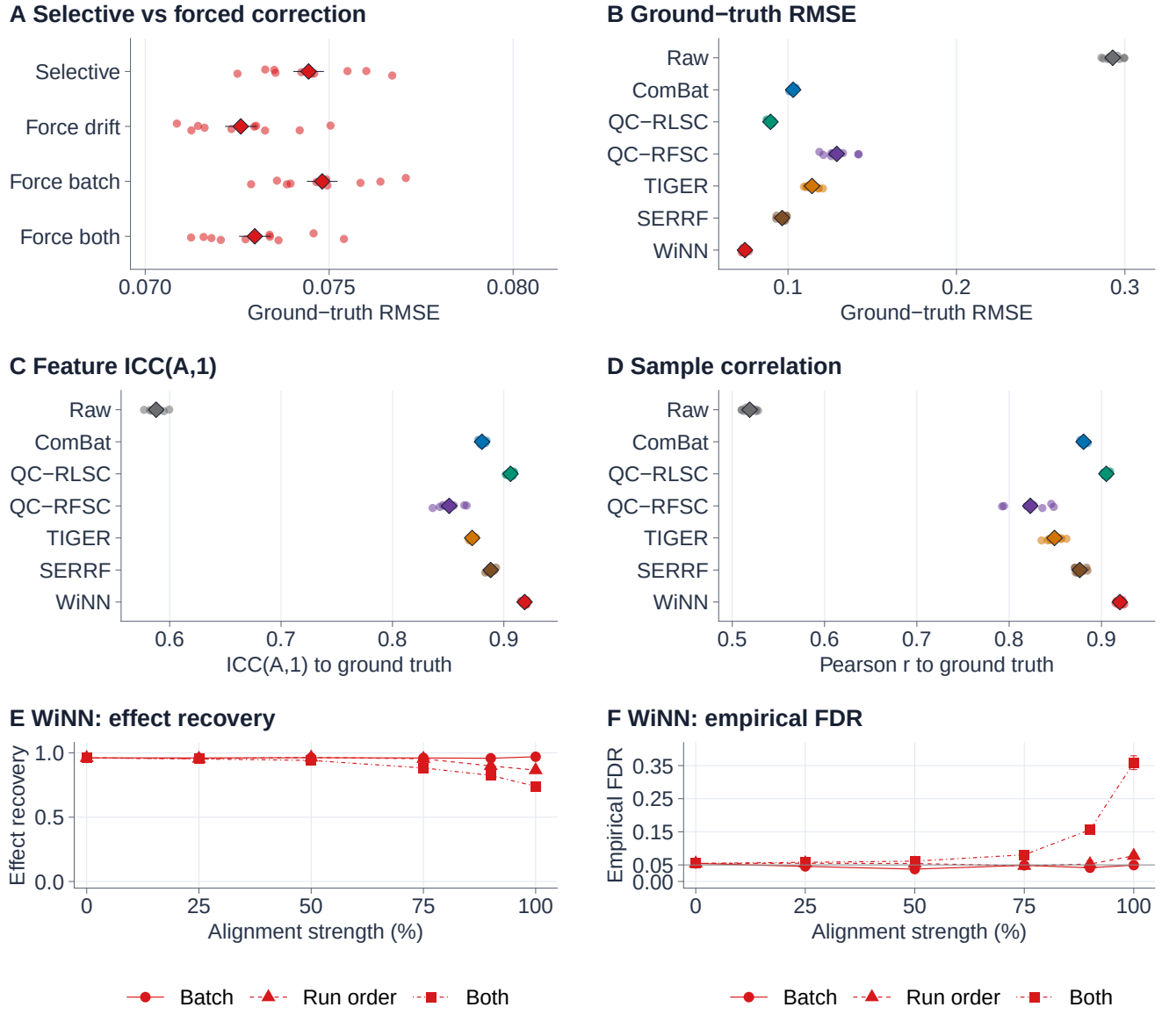
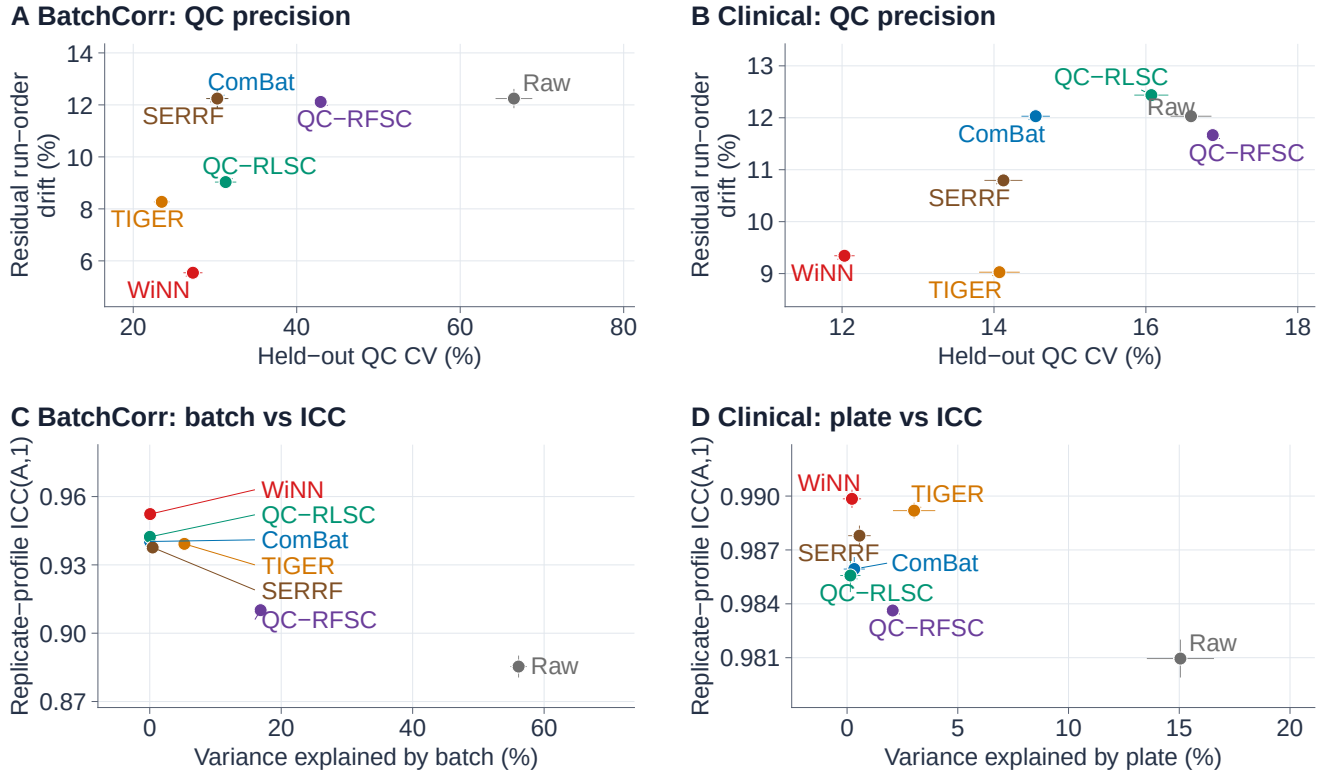
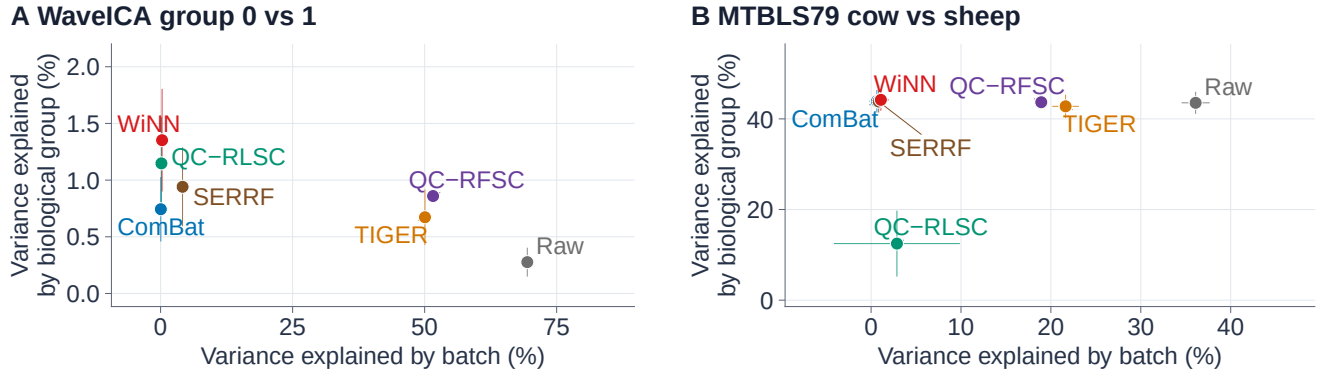

Randomized design
balanced across batches
mixed along run order
QC
colors indicate biological groups
WiNN workflow
Metabolite profiles
batch 1
batch 2
batch 3
batch 4
run order
Run-order test
GAM detrending
Batch-effect test
Batch correction
Corrected profiles


A Selective vs forced correction
Selective
Force drift
Force batch
Force both
0.070
0.075
0.080
Ground−truth RMSE
B Ground−truth RMSE
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.1
0.2
0.3
Ground−truth RMSE
C Feature ICC(A,1)
0.6
0.7
0.8
0.9
ICC(A,1) to ground truth
D Sample correlation
0.5
0.6
0.7
0.8
0.9
Pearson r to ground truth
E WiNN: effect recovery
Effect recovery
1.0
0.5
0.0
0
25
50
75
100
Alignment strength (%)
F WiNN: empirical FDR
Empirical FDR
0.35
0.25
0.15
0.05
0.00
Alignment strength (%)
Batch
Run order
Both


A BatchCorr: QC precision
Residual run−order drift (%)
Held−out QC CV (%)
ComBat
Raw
SERRF
QC−RFSC
QC−RLSC
TIGER
WiNN
B Clinical: QC precision
Residual run−order drift (%)
Held−out QC CV (%)
QC−RLSC
Raw
ComBat
QC−RFSC
SERRF
WiNN
TIGER
C BatchCorr: batch vs ICC
Replicate−profile ICC(A,1)
Variance explained by batch (%)
WiNN
QC−RLSC
ComBat
TIGER
SERRF
QC−RFSC
Raw
D Clinical: plate vs ICC
Replicate−profile ICC(A,1)
Variance explained by plate (%)
WiNN
TIGER
SERRF
ComBat
QC−RLSC
QC−RFSC
Raw


A WaveICA group 0 vs 1
Variance explained by biological group (%)
2.0
1.5
1.0
0.5
0.0
WiNN
QC−RLSC
SERRF
ComBat
QC−RFSC
TIGER
Raw
0
25
50
75
Variance explained by batch (%)
B MTBLS79 cow vs sheep
Variance explained by biological group (%)
40
20
0
WiNN
QC−RFSC
Raw
ComBat
TIGER
SERRF
QC−RLSC
0
10
20
30
40
Variance explained by batch (%)